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USER INTERFACE AND DESIGN

Experiment – 3(b)

Design a Wireflow Diagram for an E-Commerce Mobile Application. Evaluate the user flow through Login, Product Search, Cart, Checkout and Rating.

Wireflow Diagram

AIM:

The aim is to design a Wireflow Diagram for an E-Commerce Mobile Application that traces the complete user journey — from logging in, searching for products and adding them to the cart, through to checkout and submitting a rating — and to analyse how each screen and transition supports clear, intuitive, user-centred navigation.

PROCEDURE:

1. Identify all key screens in the e-commerce user journey: Sign In, Sign Up, Loading, Home, Search, Search Results, Product List, Product Details, Cart, Cart Pop-Up, Checkout, PayPal Payment, Checkout Complete, and Rate.
2. Sketch each screen as a wireframe using simple UI elements — input fields, buttons, images, icons, and navigation bars — without applying visual styling.
3. Connect the wireframes with annotated arrows to show screen transitions, indicating the user action (e.g., tap, swipe, submit) that triggers each flow.
4. Organise the wireflow into three logical phases: Login Process, Search and Add to Cart, and Checkout and Rate.
5. Review the complete diagram to ensure the flow is unbroken, logical, and covers both the primary path and key branch flows (e.g., Sign Up modal, Cart Pop-Up overlay).

OVERVIEW OF WIREFLOW DIAGRAM:

A Wireflow Diagram combines the structural detail of wireframes with the navigational clarity of a flowchart. Each screen is represented as a simplified wireframe showing its key UI components, while directional arrows between screens capture the interactions and transitions that drive the user forward. This makes wireflow diagrams particularly effective for mapping complex, multi-step user journeys such as those found in e-commerce applications, where users must move through authentication, discovery, selection, payment, and feedback stages in a coherent and intuitive sequence.

Group 1: Login Process

The login phase covers the entry point of the application. It includes the Sign In screen, the Sign Up modal overlay (for new users), and the Loading Progress screen that bridges authentication and the main app. The wireflow shows a Sign In screen with username and password fields and a login button as the primary path. A branching arrow leads to a Sign Up modal — depicted as a darkened overlay with a close (x) button — for users who do not yet have an account. Once authenticated via either path, an arrow leads to the Loading Progress screen, which displays a spinner to indicate that user data is being fetched before the app transitions automatically to the Home screen.

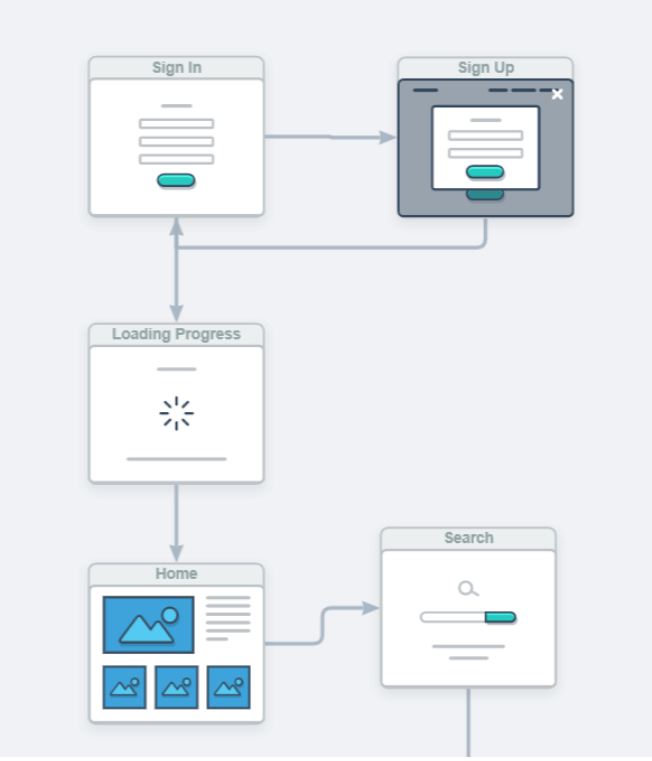

Fig 1: Login Process — Sign In → Sign Up (modal) → Loading Progress → Home

The Sign In screen uses a minimal layout to reduce friction for returning users. The Sign Up overlay pattern keeps new users contextually grounded — they remain aware they are in the authentication flow and can easily dismiss the modal to return to Sign In. The Loading screen serves as essential system feedback, preventing user confusion during data retrieval and signalling a smooth transition to the app's main experience.

Group 2: Search Product and Add to Cart

This phase covers the product discovery and selection journey. Starting from the Home screen, the user can either browse featured content or tap the search icon to navigate to the Search screen. Entering a query and submitting leads to the Search Results screen, which displays a filtered list with checkboxes and remove options. Selecting a result takes the user to the Product List screen — a two-column card grid — from which they can tap any product to reach the Product Details screen. The Product Details screen shows multiple product images, a description, and two action buttons: Add to Cart and Buy Now. Tapping Add to Cart opens a Cart Pop-Up overlay confirming the addition, before the user proceeds to the full Cart screen where all selected items are listed with remove options.

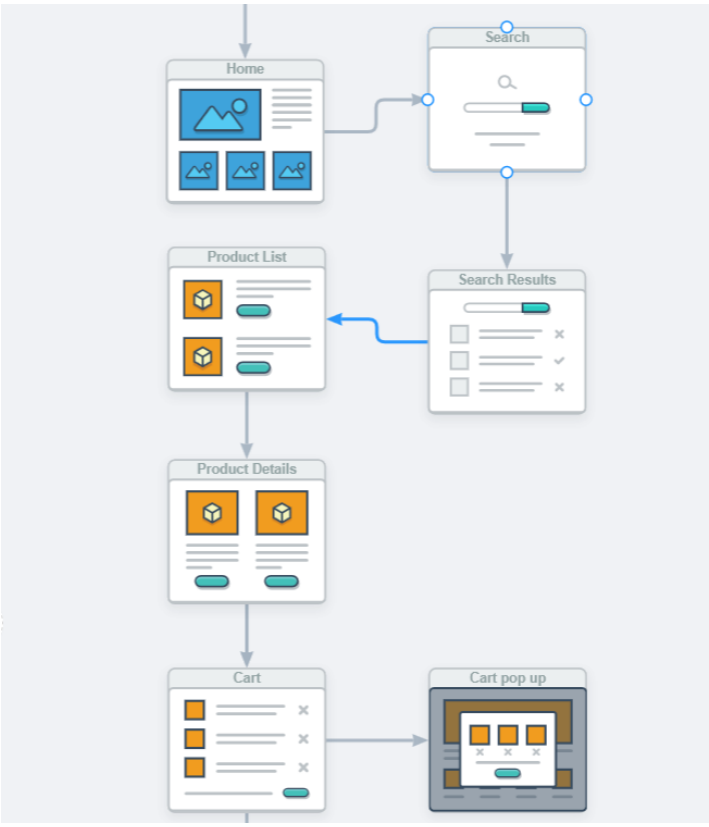

Fig 2: Search & Add to Cart — Home → Search → Results → Product List → Product Details → Cart → Cart Pop-Up

The Home screen acts as a discovery hub with a prominent banner and product thumbnails, giving users multiple entry points. The Search screen is intentionally minimal — a single active search bar — to focus attention. Search Results use checkboxes and close icons to communicate interactivity clearly. The Product Details screen's two action buttons (Add to Cart and Buy Now) cater to both deliberate and impulsive buyers. The Cart Pop-Up is a non-intrusive overlay that confirms the action without interrupting the browsing flow, reducing cognitive friction and encouraging continued product exploration.

Group 3: Checkout and Rate

The final phase takes the user from cart to completed purchase and closes the interaction loop with a rating prompt. From the Cart screen, a proceed button leads to the Checkout screen, which presents a billing summary, shipping details, and a checklist confirmation alongside a payment button. Tapping payment navigates to the PayPal screen — an external payment gateway represented as a modal with the PayPal logo and a single confirm button. On successful payment, the wireflow returns to the Checkout Complete screen, displaying a large tick icon and a success message. A continue button then leads to the Rate screen, where the user is presented with an interactive five-star widget and an optional comment field, followed by a submit button to finalise their feedback.

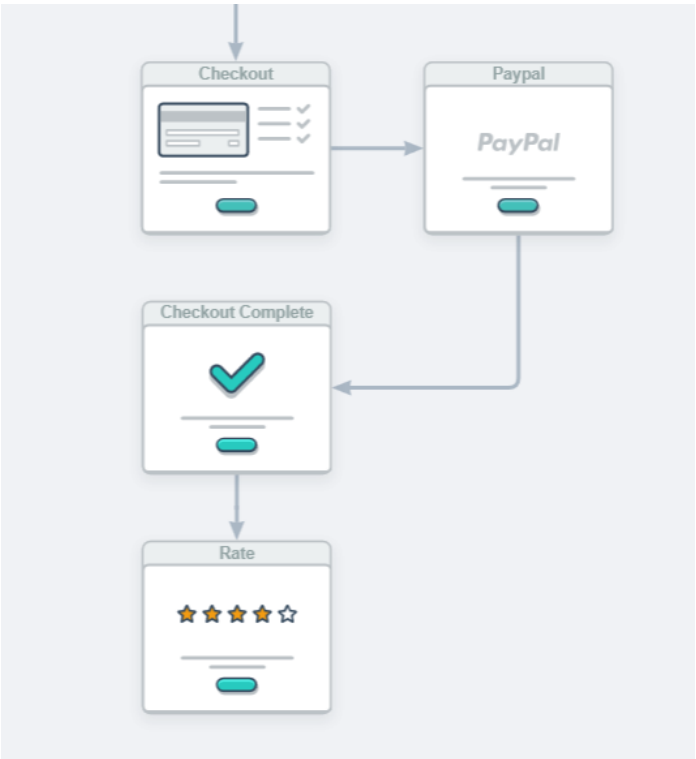

Fig 3: Checkout & Rate — Cart → Checkout → PayPal → Checkout Complete → Rate

The Checkout screen uses a checklist design to instil confidence before payment — users can visually verify their order is correct before committing. The PayPal screen's recognisable branding and single-action layout reinforce trust and security. The Checkout Complete screen uses a universally understood tick icon to provide immediate, unambiguous success feedback, reducing post-purchase anxiety. Placing the Rate screen immediately after checkout capitalises on the positive emotional state following a successful transaction, increasing the likelihood of users submitting genuine and favourable feedback.

ANALYSIS:

- The wireflow clearly maps a logical, unbroken user journey from Sign In to Rating, covering all primary and secondary interaction paths.
- Modal overlays for Sign Up and Cart Pop-Up are effective design choices — they provide necessary functionality without breaking the user's sense of context or flow.
- The three-phase structure (Login, Discovery, Checkout) mirrors the natural mental model of an e-commerce user, reducing cognitive load at each stage.
- Familiar navigation patterns — bottom tab bar, search bar, product cards, and labeled buttons — ensure the wireflow is accessible and learnable across all user groups.
- The PayPal integration screen demonstrates how third-party trust signals can be embedded within a wireflow to address security concerns at the critical payment step.
- The Rating screen at the end of the flow closes the feedback loop, ensuring the application captures user sentiment at the point of highest satisfaction.

Complete Wireflow Diagram

The diagram below presents the full wireflow for the e-commerce mobile application, combining all three phases into a single end-to-end view. Each screen is shown as a wireframe and connected by annotated arrows indicating the user action that triggers the transition.

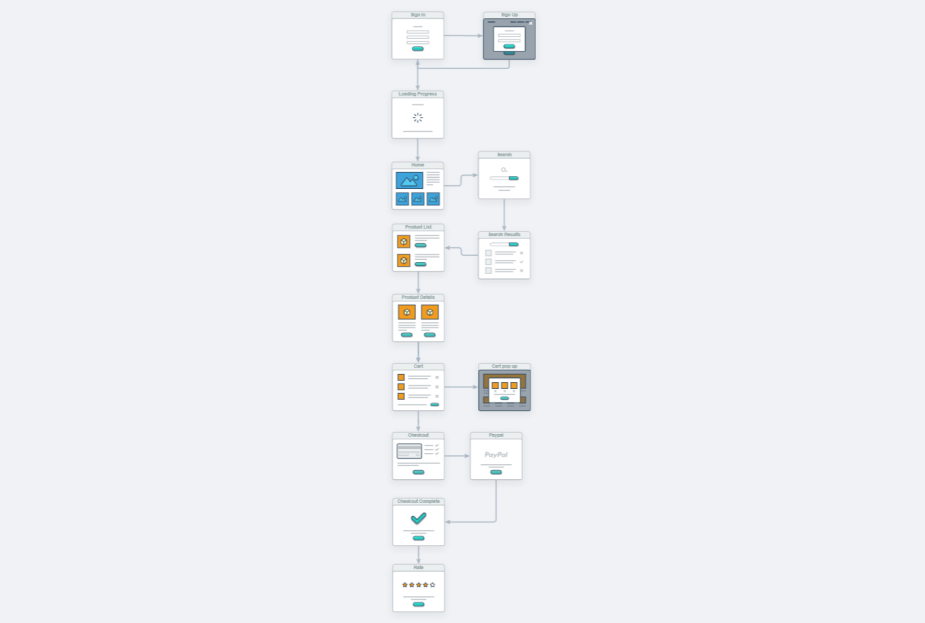

Fig 4: Complete Wireflow Diagram — Full E-Commerce User Journey (Sign In → Sign Up → Loading → Home → Search → Results → Product List → Product Details → Cart → Cart Pop-Up → Checkout → PayPal → Checkout Complete → Rate)